Analytics to Action

Exploratory Data Visualization

Anders Koed Madsen, March, 2026

Theme 2: Making Data Valuable

How to make data and data analysis valuable to different stakeholders in practice.

Three sessions

1. Datafication
2. **Exploratory data analysis**
3. Participatory data design

Readings for today

- Venturini, T., Jacomy, M., & Jensen, P. (2021). What do we see when we look at networks: Visual network analysis, relational ambiguity, and force-directed layouts.
- Sapienza, A., & Lehmann, S. (2021). A view from data science.

Agenda for this lecture

- Exploratory Data Analysis (EDA)
- Why does visualization matter?
- Visual network analysis (VNA) as a case of EDA
- Exercise in class: VNA in practice
- Interview with Johan on EDA in Will & Agency
- Prompt for your case presentations

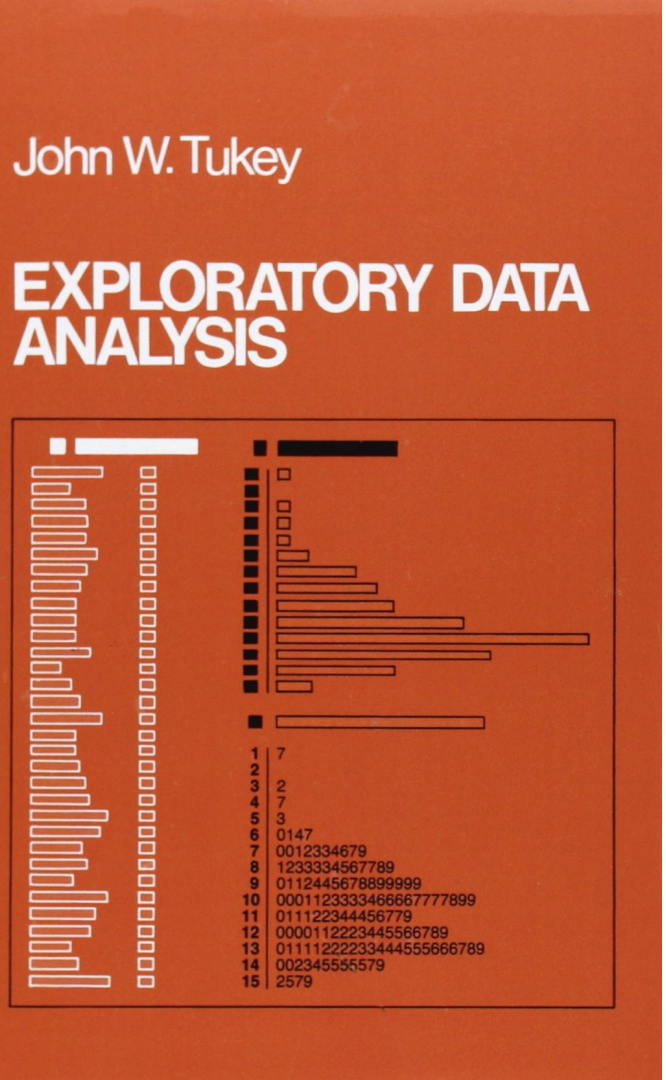
EDA at a glance

*"The main purpose of EDA is to help look at data **before making any assumptions**. It can help identify obvious **errors**, as well as better understand **patterns** within the data, detect **outliers or anomalous events**, find **interesting relations** among the variables."*

— IBM Think Blog

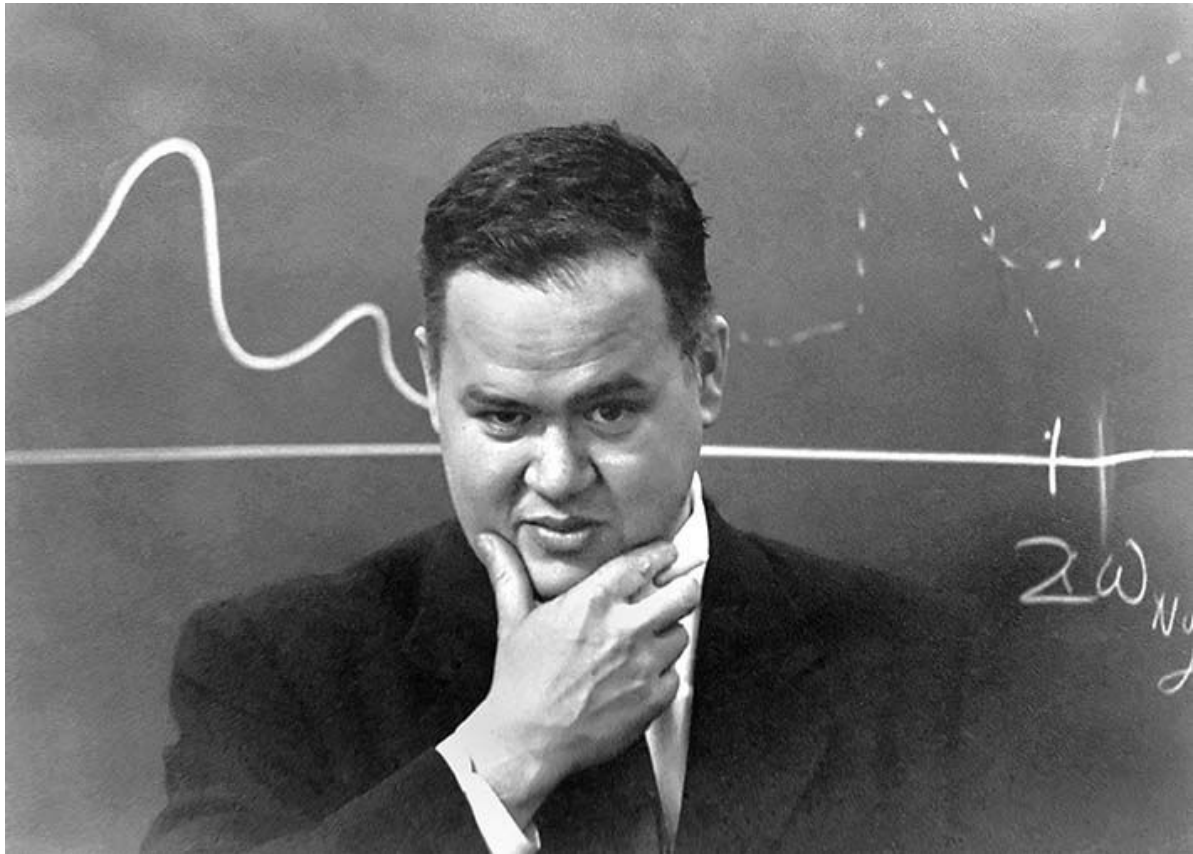
*"In essence, it involves thoroughly examining and characterizing your data in order to find its underlying **characteristics**, possible **anomalies**, and hidden **patterns and relationships**."*

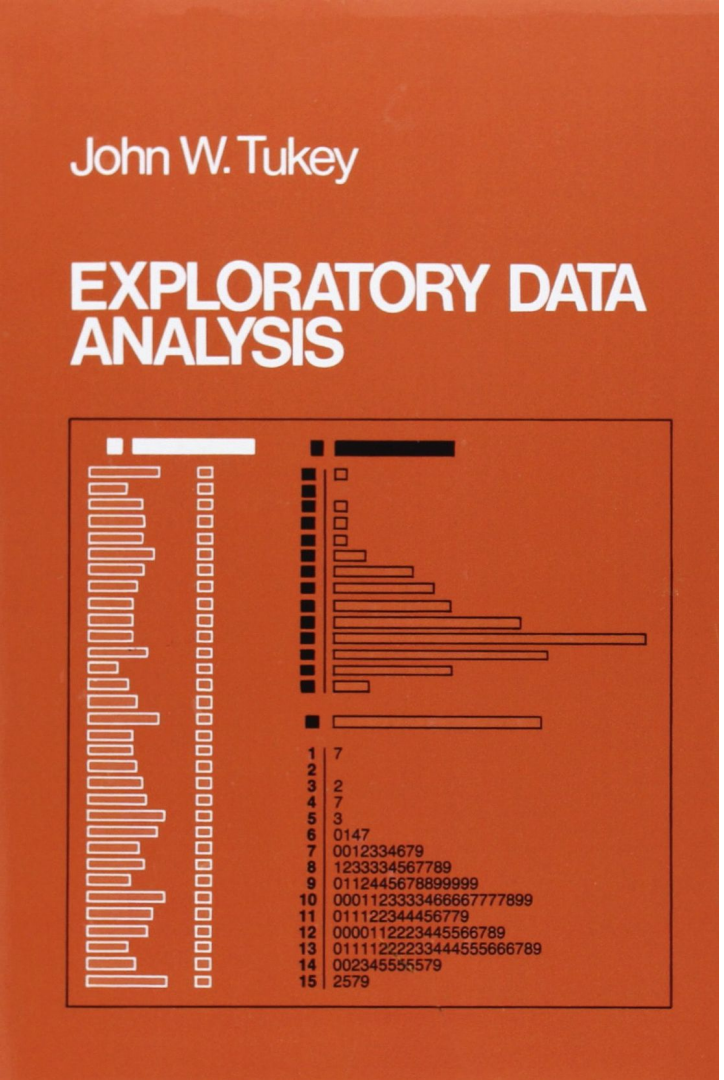
— Towards Data Science



◀ Wrote the book on Exploratory Data Analysis in 1977

John Tukey, American statistician/mathematician, inventor of the box plot, professor at Princeton▼





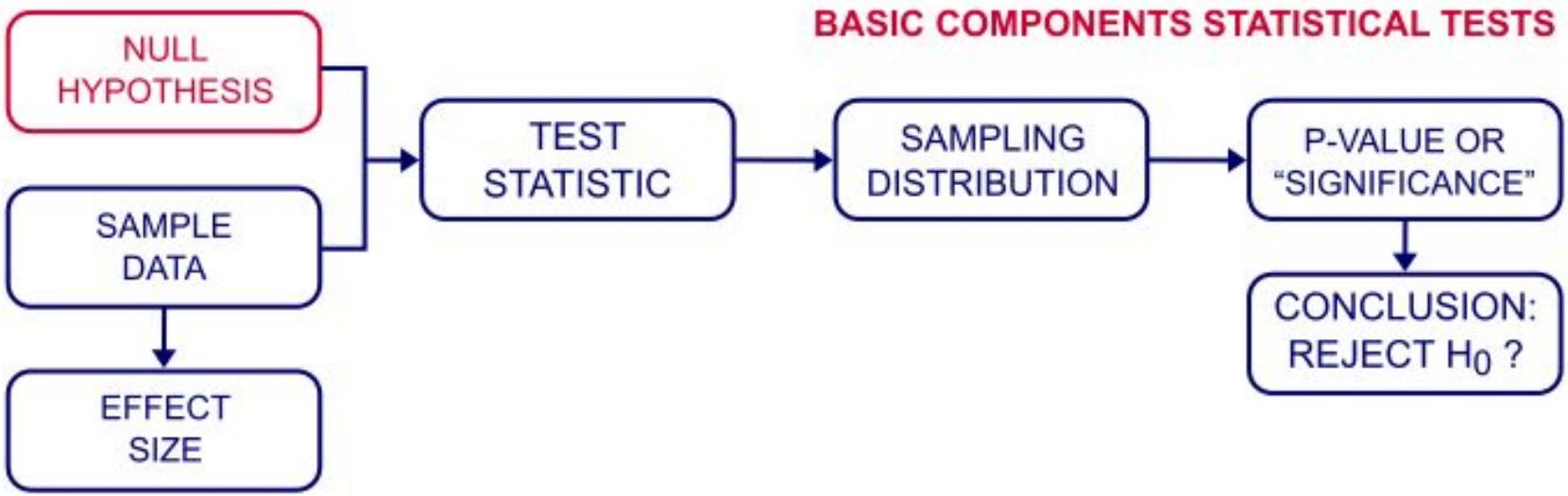
"The greatest value of a picture is when it ***forces us to notice*** *what we never expected to see."*

— Tukey, 1977

"Far better an approximate answer to ***the right question****, which is often vague, than an exact answer to the wrong question, which can always be made precise."*

— Tukey, 1977

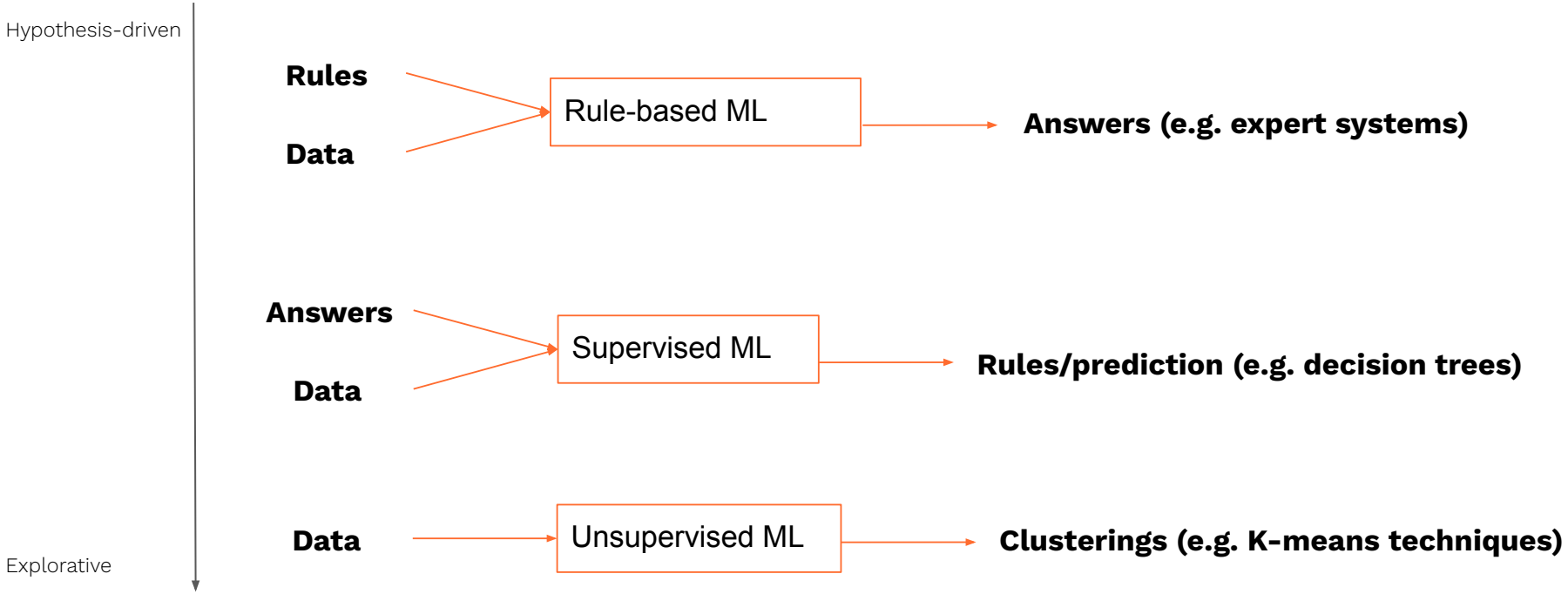
Reaction to hypothesis-driven statistics



A clear example of this experimental design is the practice of preregistration, which is becoming predominant especially in psychology, economics, and political science.

When research is preregistered, not only the questions/ hypotheses need to be specified but also the variables and tests (Sapienza and Lehmann, 2021)

[data scientists] are not hypothesis-driven in the sense of many social scientists. As we explore the data, we form informal theories of what might be driving the patterns we see. We learn what the right questions (and variables) are (Sapienza and Lehmann, 2021)


Hypothesis-driven
Rules
Data
Rule-based ML
Answers (e.g. expert systems)
Answers
Data
Supervised ML
Rules/prediction (e.g. decision trees)
Data
Unsupervised ML
Clusterings (e.g. K-means techniques)
Explorative

Are You Creditworthy? The Algorithm Will Decide

Whether we ought to have faith in algorithmic credit scoring is hard to answer, given the impenetrability of machine learning.

Question for discussion (5 minutes)

Think of your case. Can you think of both an explorative and hypothesis-based approach to analyzing the data you will be working with.

- Please describe in a bit of detail what you would do with the data.

  (e.g. a hypothesis driven approach to Rigshopitalet would be to start testing the hypothesis that people above 40 are more prone to show up than younger people)

DTU

Join at:
vevox.app

ID:
148-629-575

Why does visualization matter?

I		II		III		IV	
x	*y*	*x*	*y*	*x*	*y*	*x*	*y*
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

▲Francis Anscombe, British statistician, professor at Yale

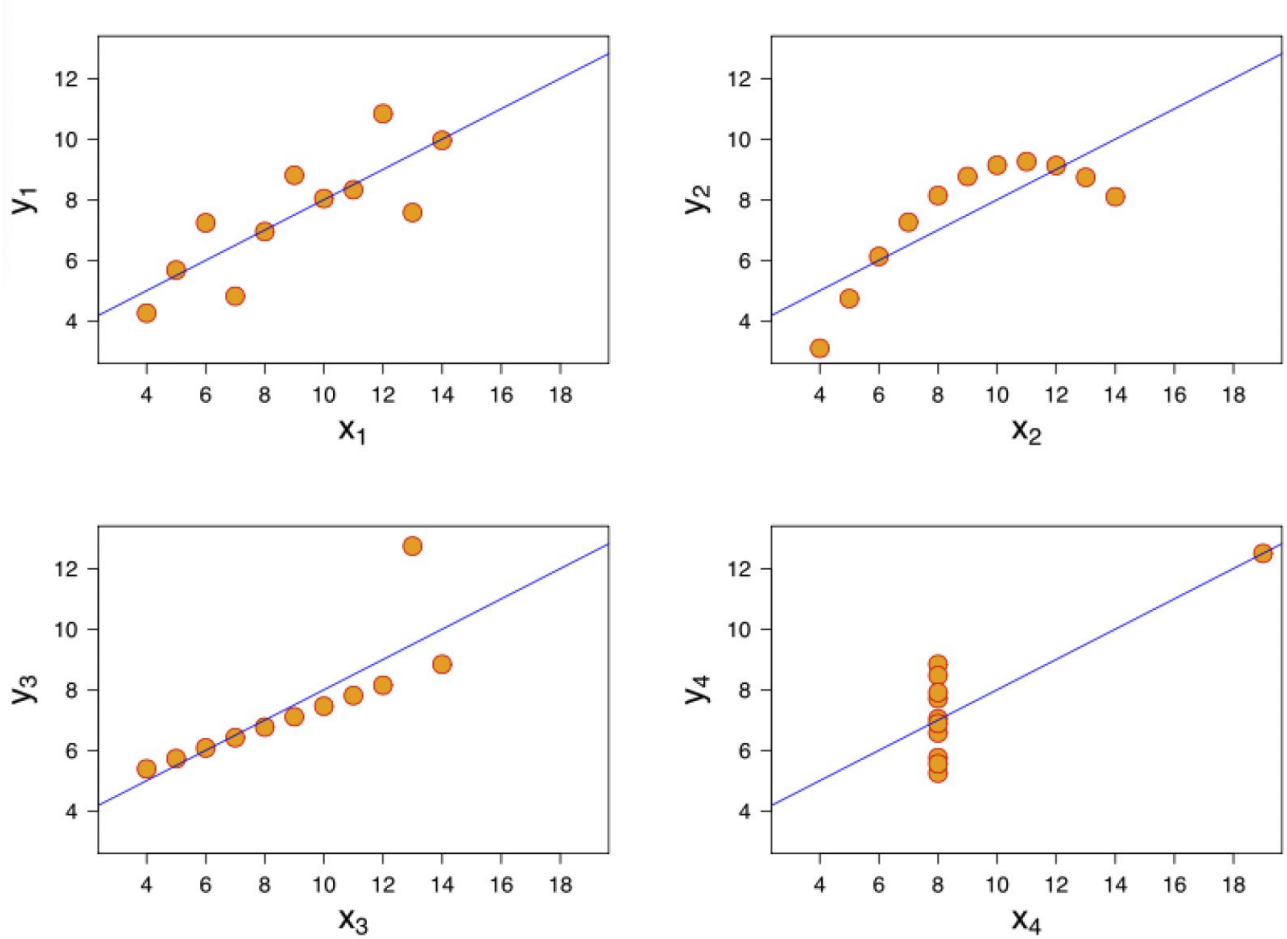
◀ **Anscombe's quartet** (1973)

When metrics fall short…

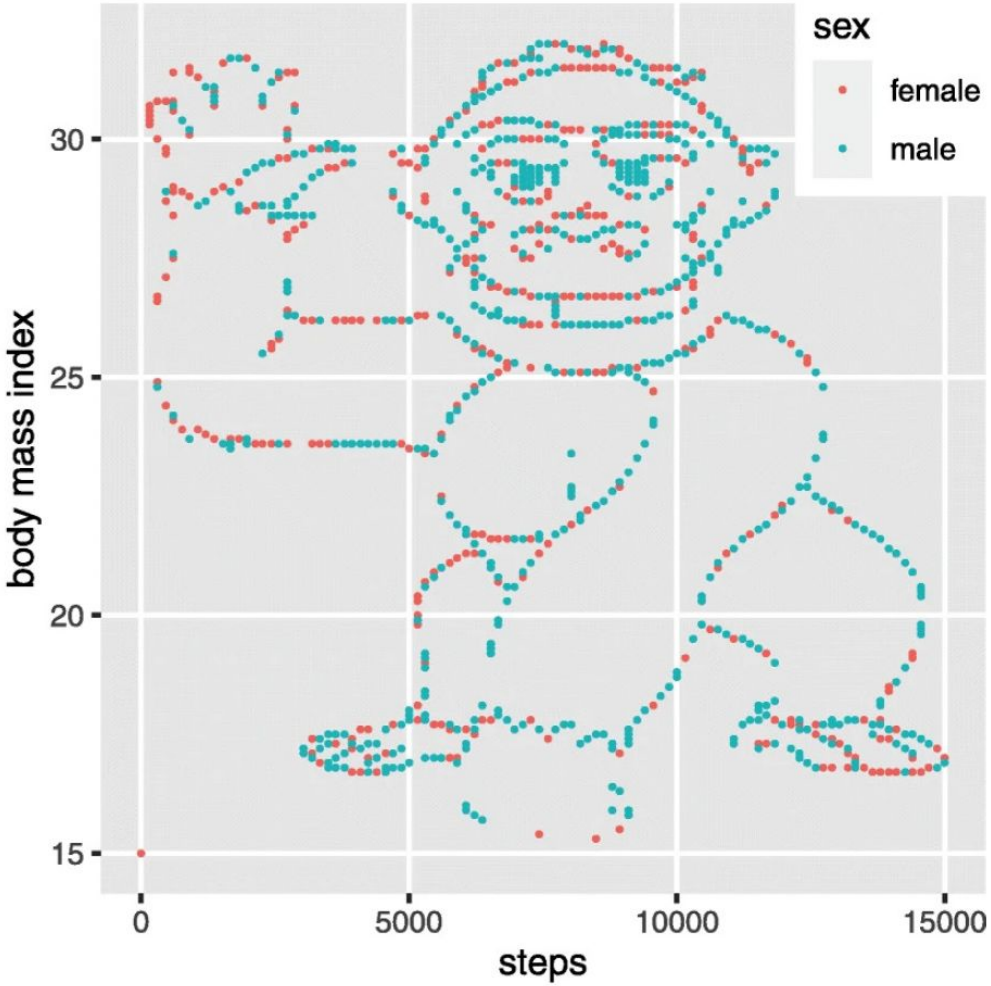
Statistic	Dataset I	Dataset II	Dataset III	Dataset IV
Mean of x	9.00	9.00	9.00	9.00
Mean of y	7.5009	7.5009	7.5000	7.5009
Variance of x	11.00	11.00	11.00	11.00
Variance of y	4.1273	4.1276	4.1226	4.1232
Correlation (x,y)	0.81642	0.81624	0.81629	0.81652
Regression slope	0.50009	0.50000	0.49973	0.49991
Regression intercept	3.00009	3.00091	3.00245	3.00173

▲ Anscombe's quartet

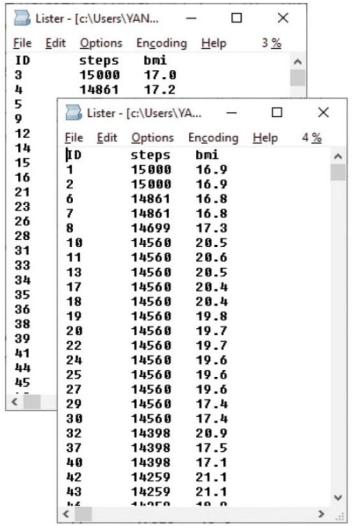
…visualization can be the quickest way



◀ Anscombe's quartet



Artificial dataset given to students with and without hypothesis on the relationship between BMI and steps taken ▸



◀ Plot of the data

Contingency table for students with and without hypothesis ▼

	Gorilla <u>not</u> discovered	Gorilla discovered
Hypothesis-focused	14	5
Hypothesis-free	5	9

Yanai, I., Lercher, M. (2020) **A hypothesis is a liability**. Genome Biol 21, 231

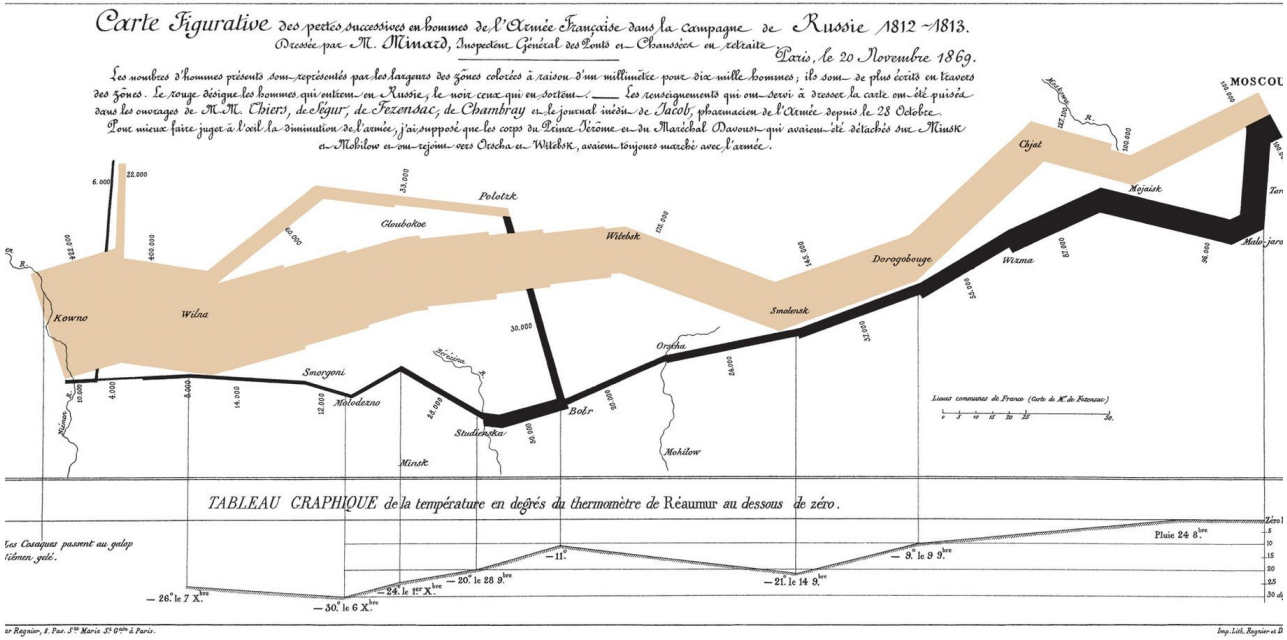
“The Invisible Gorilla”

When people focus on a specific task, they may become *psychologically blind* to unexpected features of the world

When people focus on testing a specific hypothesis, they may become *psychologically blind* to unexpected features of the data

Not to be confused with *explanatory* visualization

Where you curate the visualization to convey a point



◀ **Charles Minard's** map of Napoleon's 1812 campaign in Russia

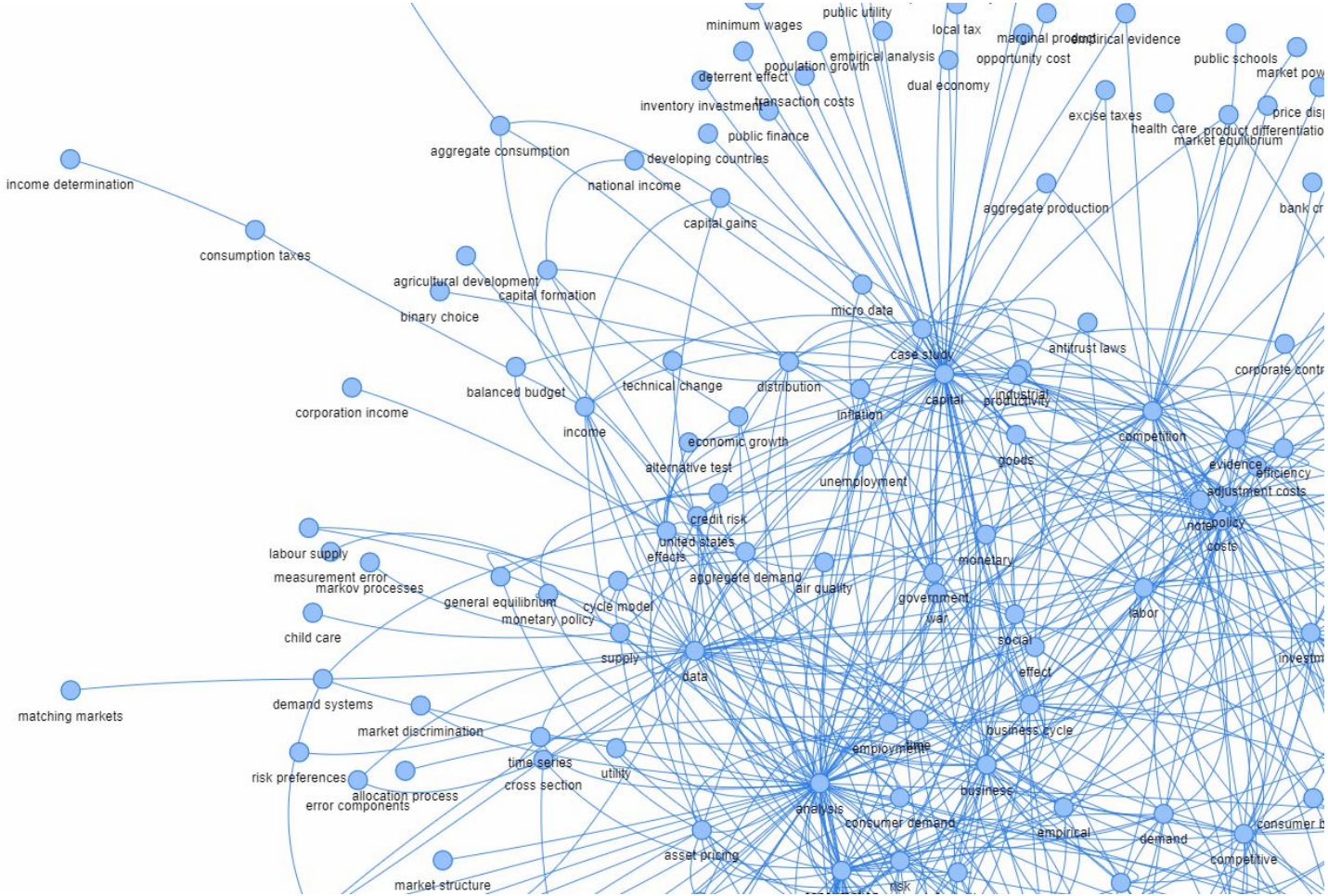
Question for discussion (5 minutes)

Please note down types of data visualizations and indicate in parentheses if you think of it as suitable for either explorative, hypothesis driven or both:

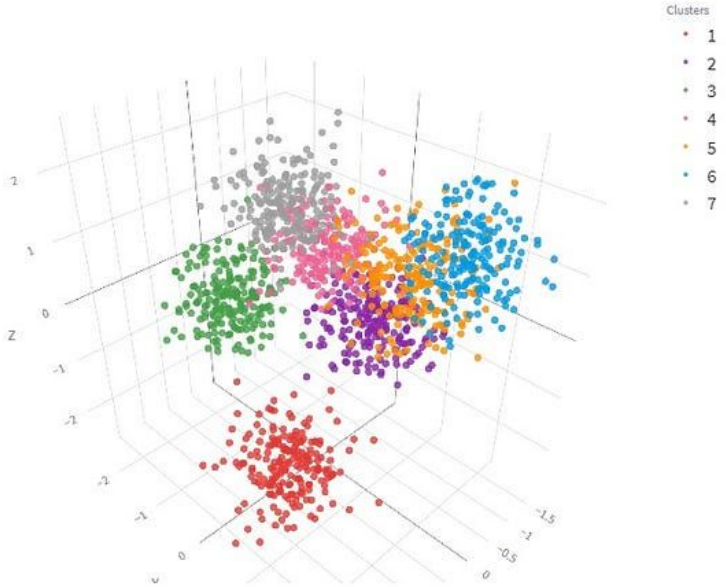
- e.g. Bar charts (hypothesis driven because they require pre-defined categories)

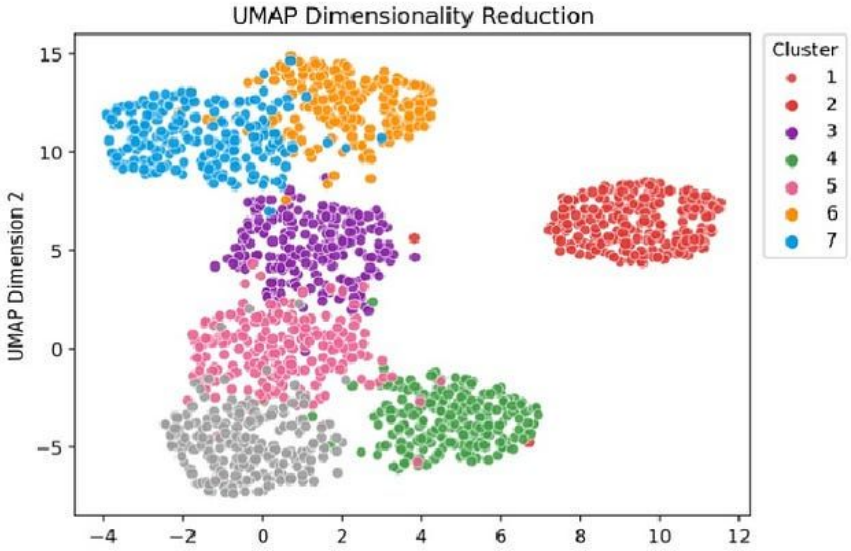
Visual Network Analysis

A paradigmatic case of EDA


minimum wages
public utility
local tax
marginal product
empirical evidence
empirical analysis
opportunity cost
public schools
population growth
deterrent effect
dual economy
inventory investment
transaction costs
excise taxes
health care
product differentiatio
market equilibrium
public finance
aggregate consumption
developing countries
income determination
national income
aggregate production
capital gains
consumption taxes
agricultural development
capital formation
binary choice
micro data
antitrust laws
case study
balanced budget
technical change
distribution
corporation income
capital
industrial productivity
inflation
income
competition
economic growth
goods
alternative test
unemployment
evidence
efficiency
adjustment costs
credit risk
policy
note
costs
labour supply
united states
effects
monetary
measurement error
markov processes
aggregate demand
air quality
general equilibrium
cycle model
government
monetary policy
war
labor
child care
social
supply
effect
data
matching markets
demand systems
market discrimination
business cycle
employment
time series
utility
risk preferences
cross section
allocation process
error components
business
analysis
consumer demand
empirical
demand
asset pricing
competitive
market structure
risk

Point cloud in a 3-dimensional space





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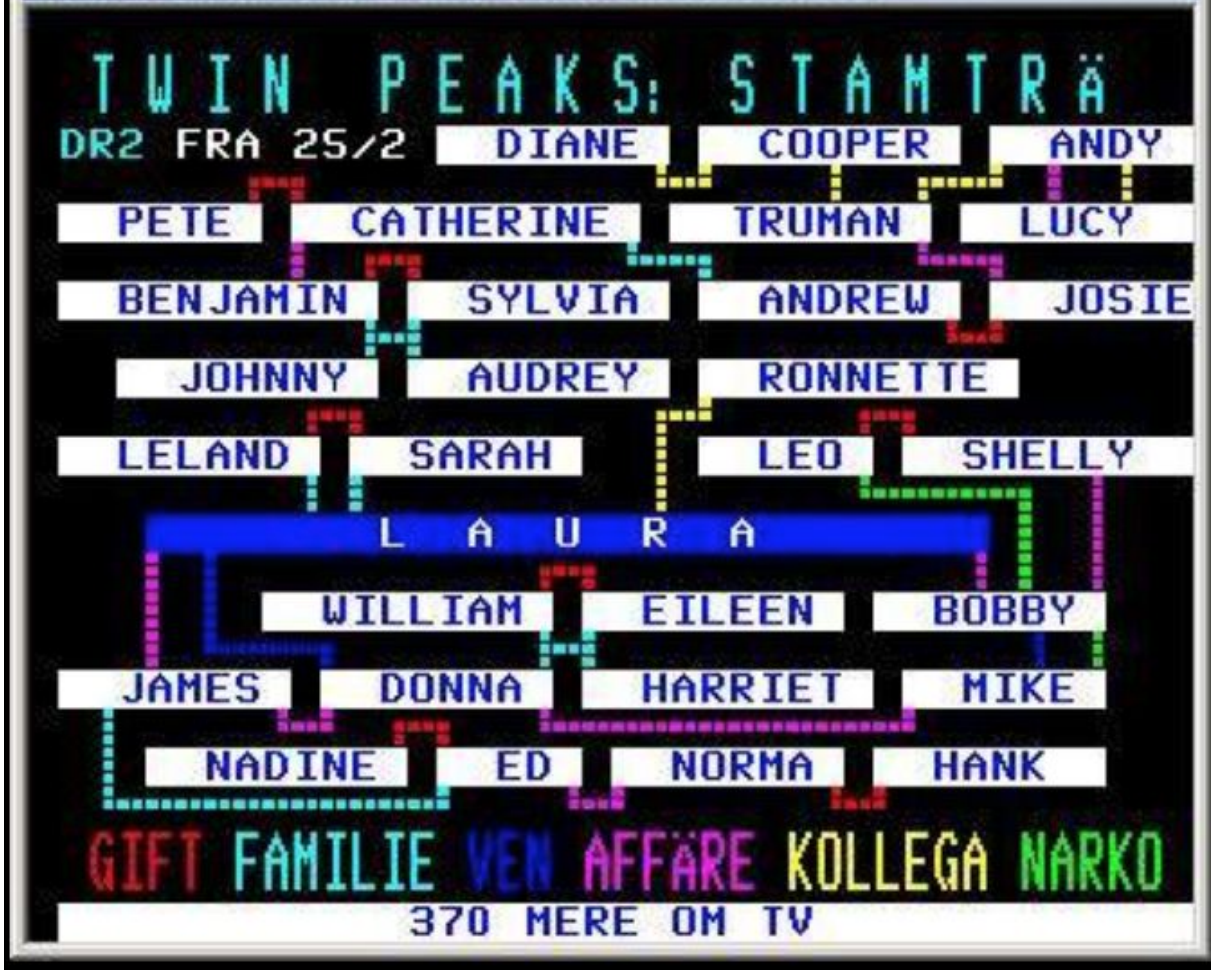
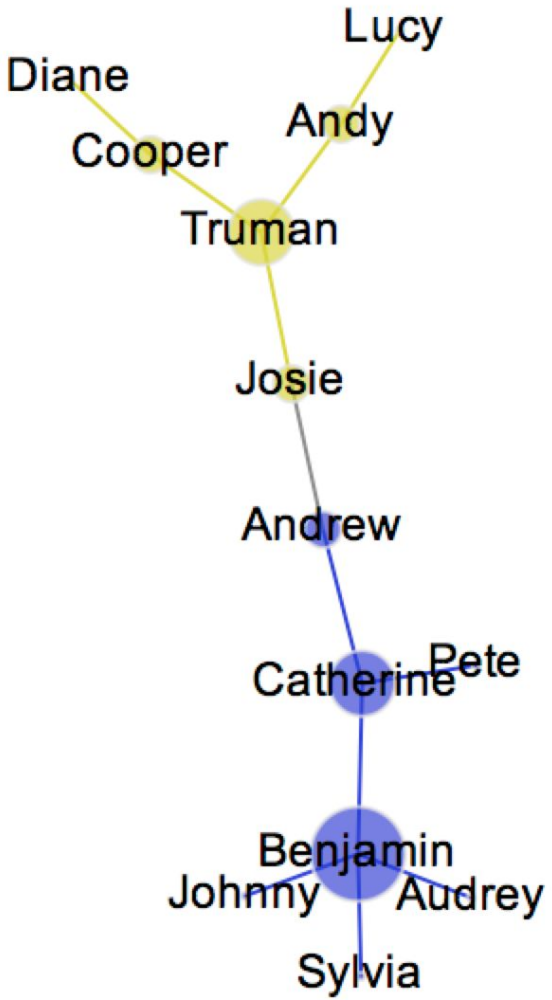
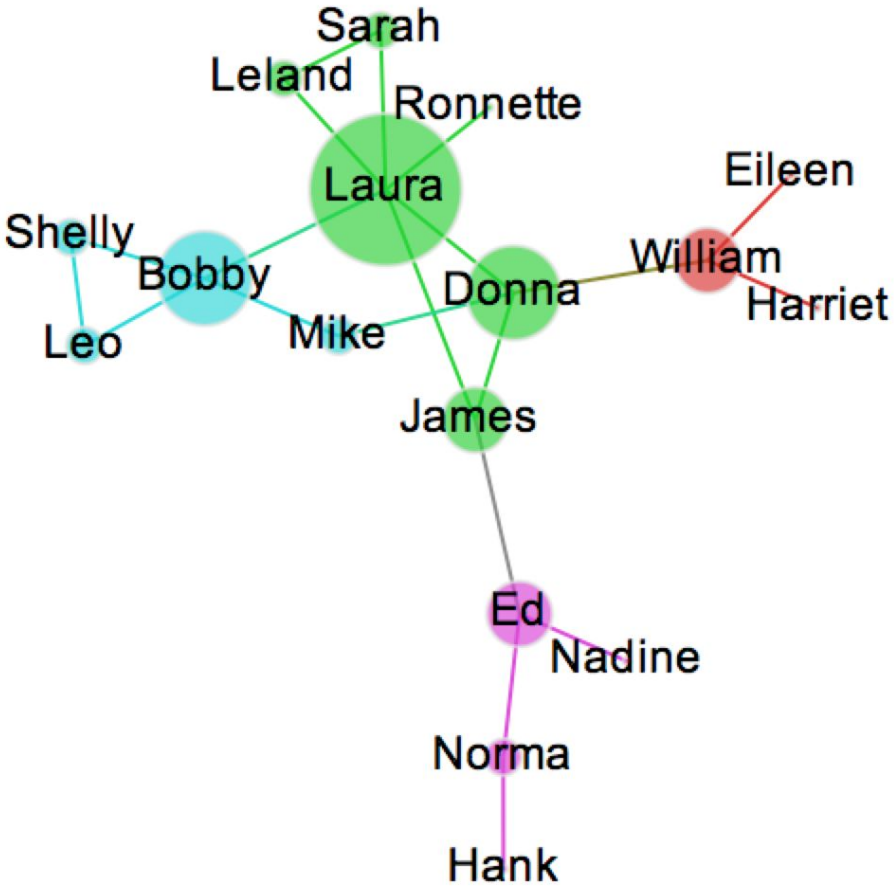
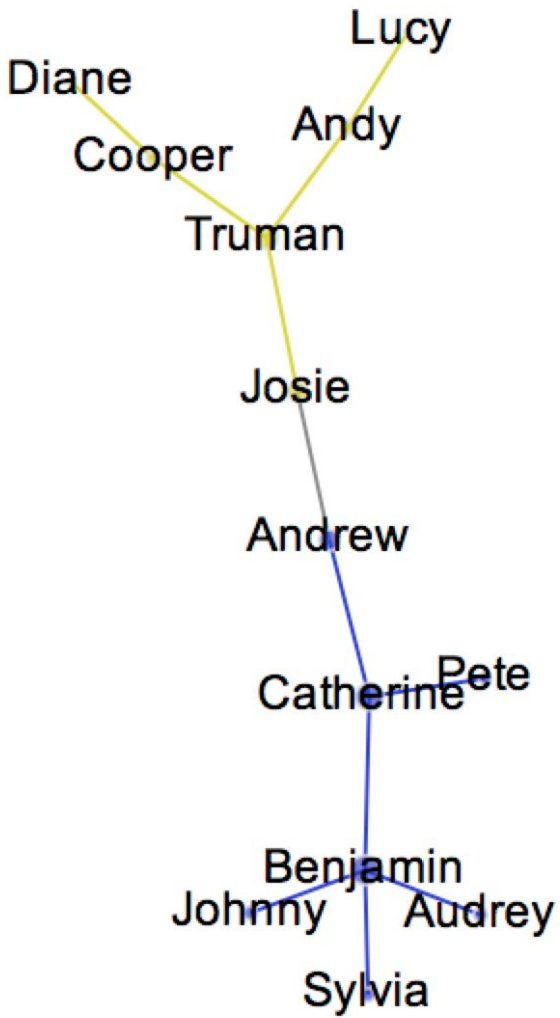
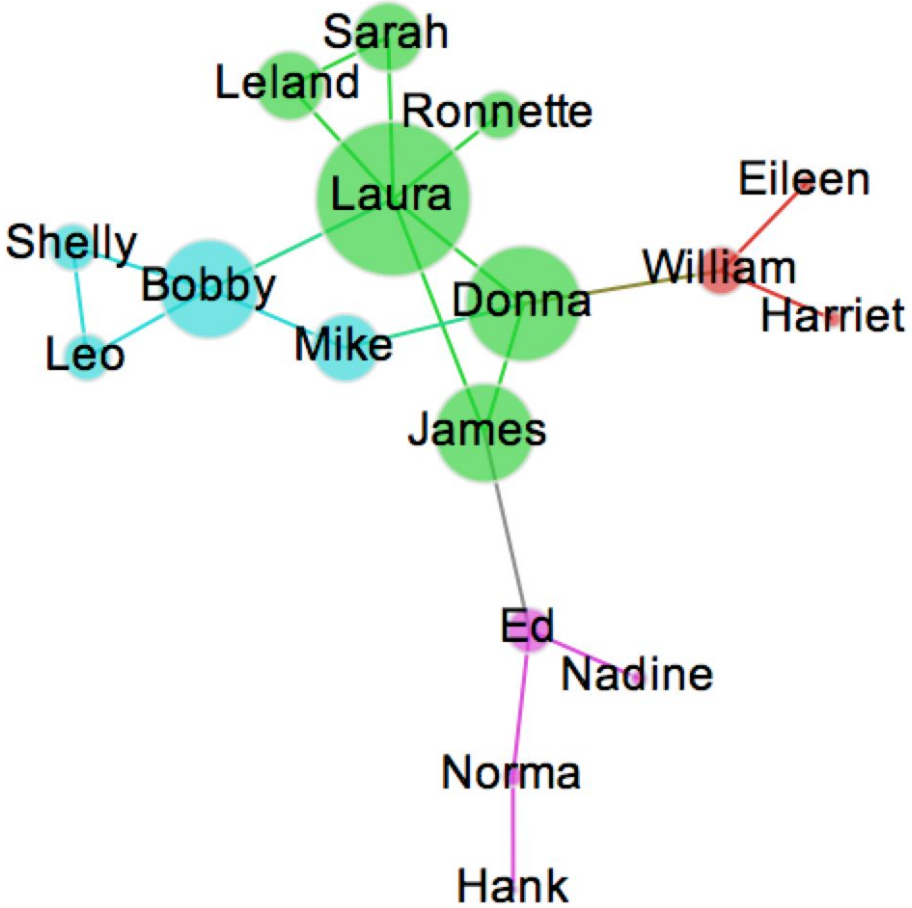




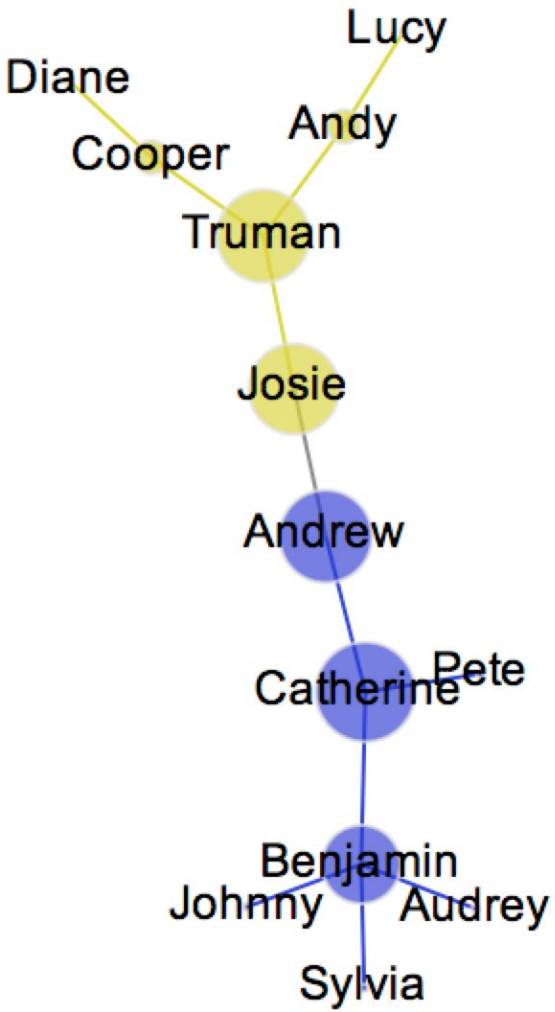


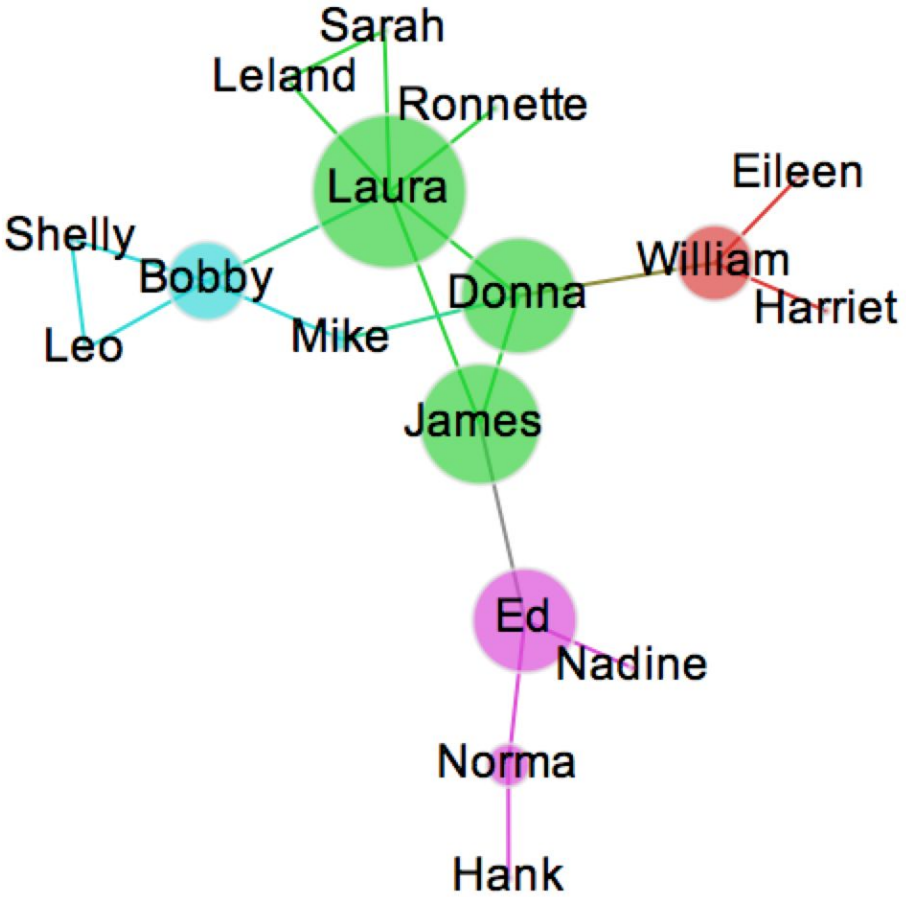
Layout: ForceAtlas2 (force directed)
Colour: Louvain Modularity
Size: Degree





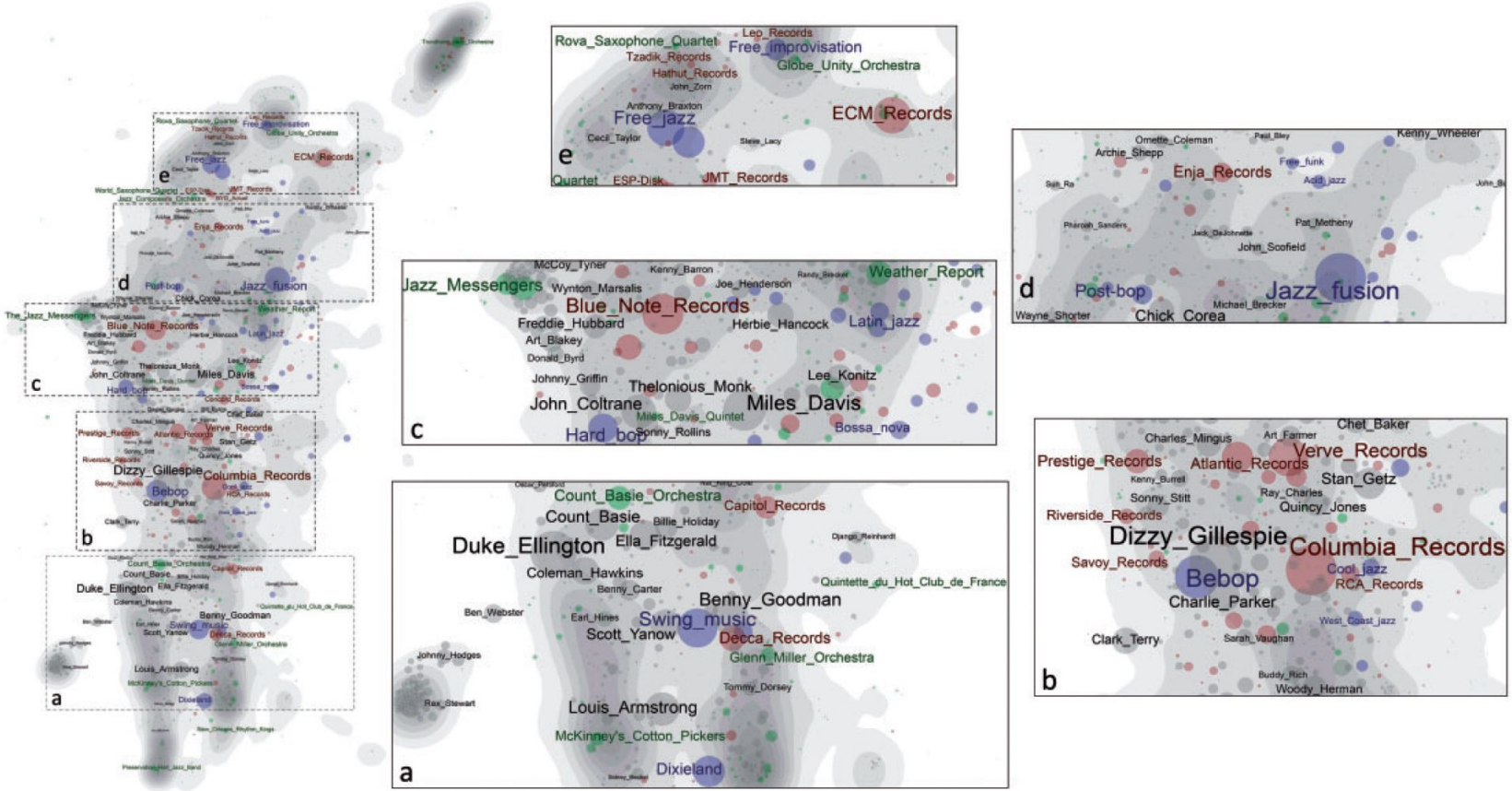
Layout: ForceAtlas2 (force directed)
Colour: Louvain Modularity
Size: Eigen Vector Centrality





Layout: ForceAtlas2 (force directed)
Colour: Louvain Modularity
Size: Betweenness

Forming hypotheses about music on Wikipedia through exploration of network topology



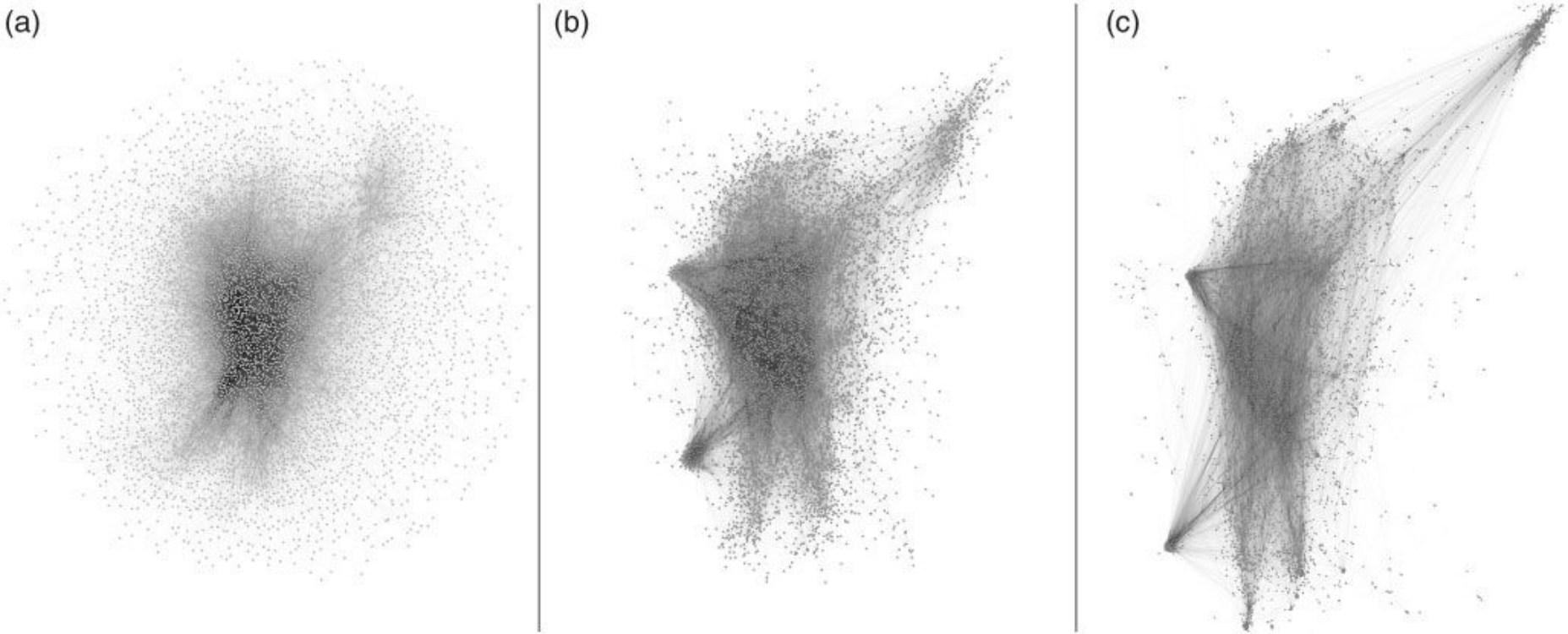


Figure 3. The "jazz network" spatialized (a) with the algorithm proposed by Fruchterman and Reingold (1991), (b) with ForceAtlas2 (with default parameters) and (c) with ForceAtlas2 with tweaked parameters for LinLog mode and gravity. This and all images created for this paper are available at: https://github.com/tommv/ForceDirectedLayouts.

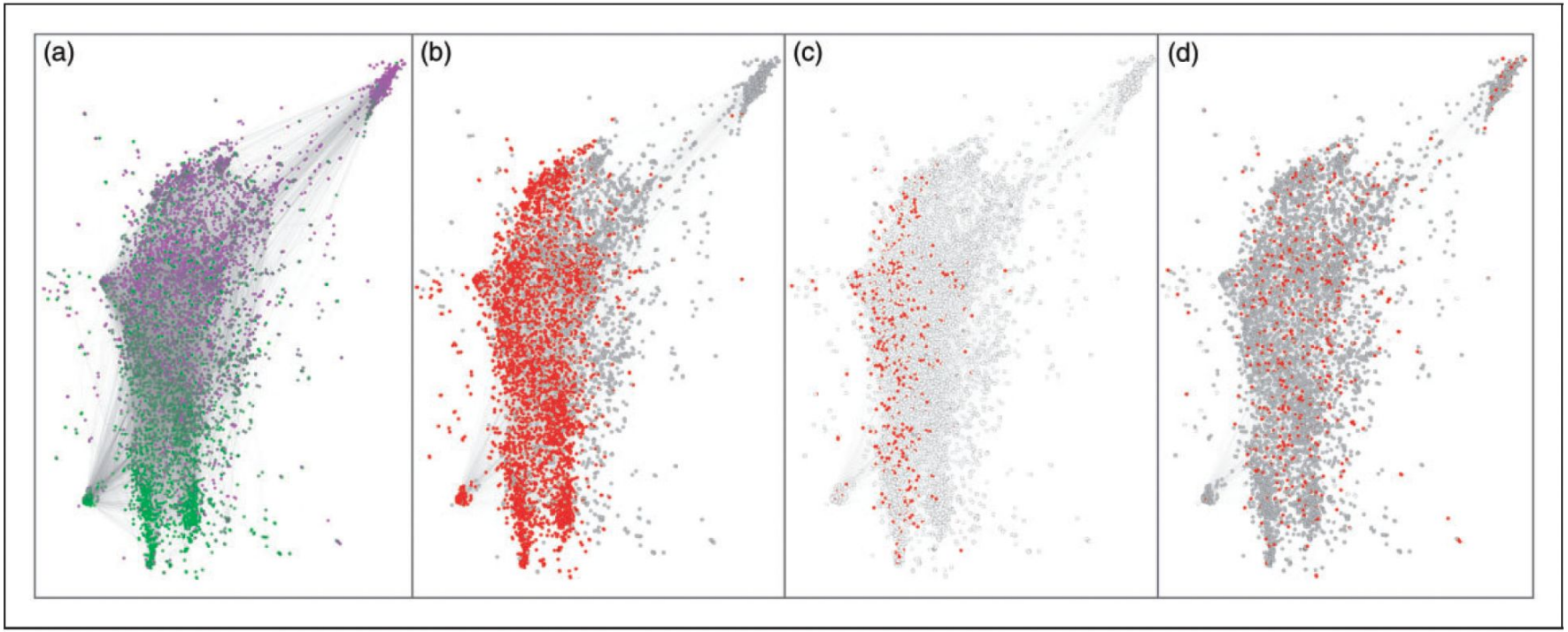
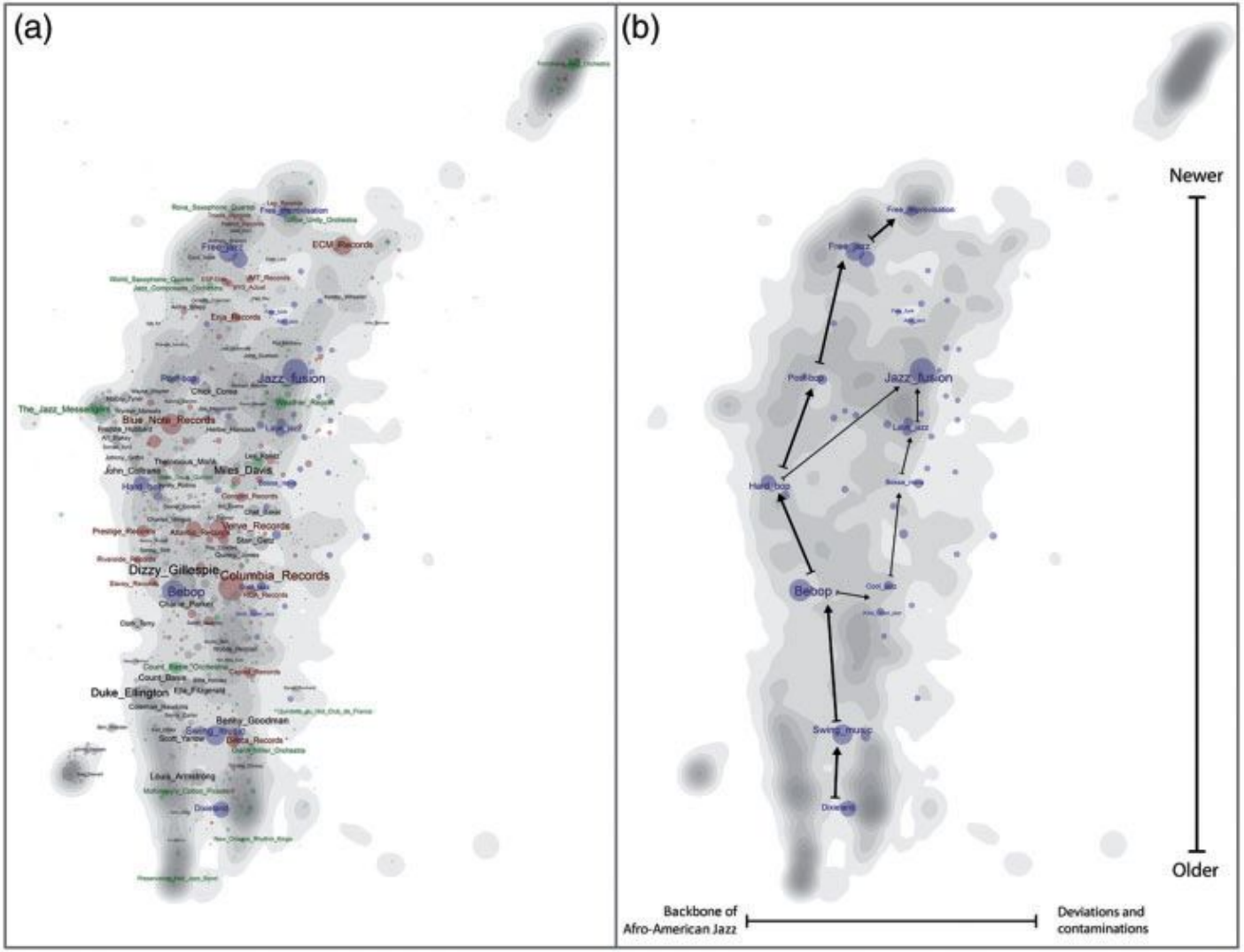
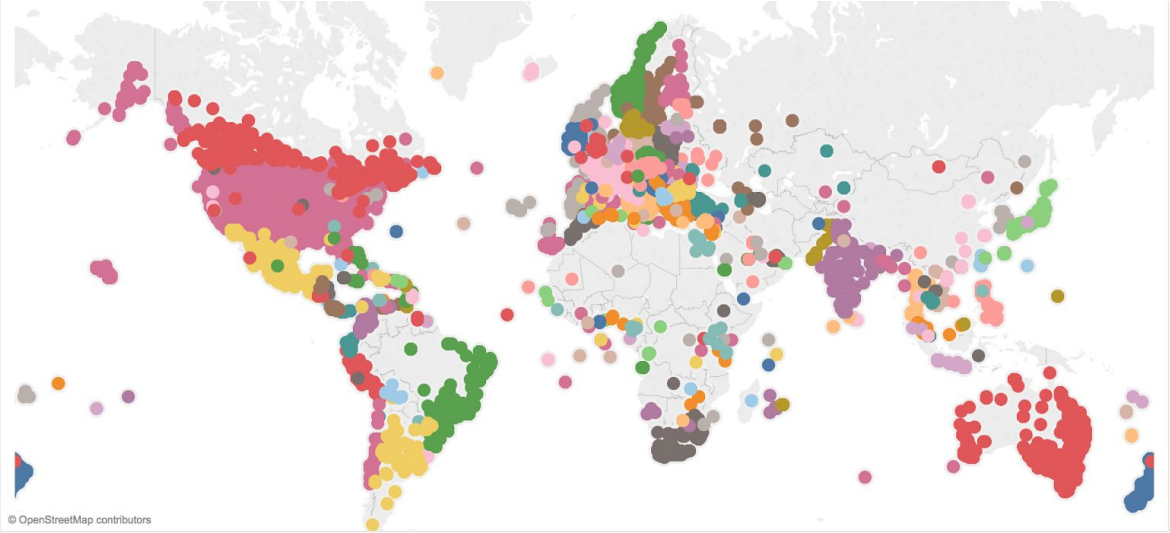


Figure 5. The "jazz network" with nodes colored according to: (a) the year of their birth or inception (from green for earliest dates to magenta for most recent); (b) their nationality (red for US, gray for all other countries, white for not available); (c) their ethnic groups (red for African American, gray for other ethnic groups, white for not available); and (d) their gender (red for women, gray for men, white for not available or others).


(a)
Free_Jazz
ECM_Records
Enja_Records
Jazz_fusion
Blue_Note_Records
Miles_Davis
John_Coltrane
Verve_Records
Dizzy_Gillespie
Columbia_Records
Bebop
Duke_Ellington
Benny_Goodman
Louis_Armstrong
(b)
Newer
Older
Free_Jazz
Postbop
Jazz_fusion
Hard_bop
Bebop
Swing_music
Backbone of Afro-American Jazz
Deviations and contaminations

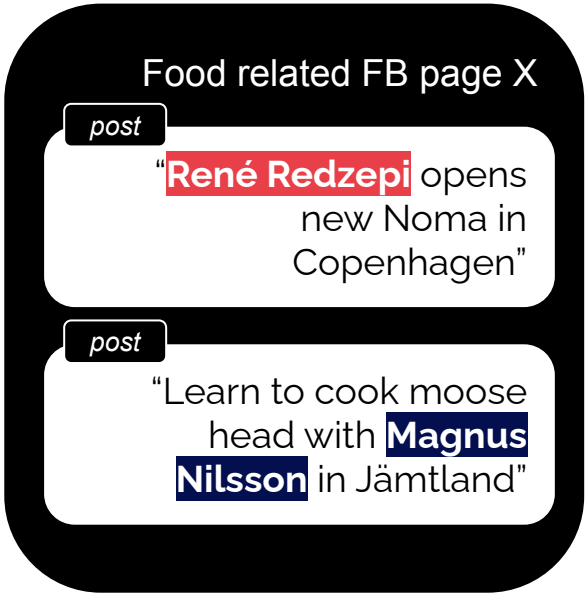
Exercise

How are chefs talked about on food-related Facebook? Form some hypotheses through VNA.



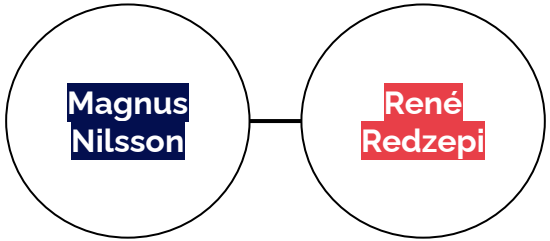
Dataset: All posts (text) from 242.000 food-related Facebook pages worldwide between 2010 and 2018

+ 670 names of famous chefs (scraped from Wikipedia)



NETWORK

Nodes = chef names
Edges = co-mention on food-related page



Variables on nodes

- **Country names:** To what extent has a chef been talked about by pages from this country.
- **Nordic:** Is the chef from one of the nordic countries? If yes, which?
- **Mention count:** How many times is the chef mentioned?
- **Page count:** How many different pages are mentioning this chef?
- **Country count:** How may different countries are the pages mentioning this chef from?
- **Degree:** How many chefs other chefs is this chef mentioned together with?
- You can also use the statistics module to compute the modularity class (using the Louvain method), i.e. cluster the chefs based on the topology of the network.

Variables on edges

- **Raw overlap:** How many pages are co-mentioning two chefs?
- **Normalized overlap:** How many pages are co-mentioning two chefs as a proportion of the max possible given as the number of pages mentioning each chef?
- **Only nordic overlaps:** If only pages from the nordic countries are taken into account as co-mentioning chefs.

Let’s try to enact an EDA process on this data together….

Engage in EDA yourself (7 minutes)

Download the .gexf file from Learn

Analyse it in Gephi Lite: **https://gephi.org/gephi-lite/**

EDA at Will & Agency

Conversation with Johan

Prompts for your case work and presentations

- To what extent do you have clear hypotheses? To what extent are they testable?
- What would be good exploratory visualizations to guide you towards more data-driven hypotheses?
- Is your data relational? How could you explore it through visual network analysis? Concretely: what would be the nodes, what would be the edges?